~~2.20~~

2.20. $F = m\mu r$ 由牛顿第二定律 $F = ma$.

故 $a = \mu r$.

由 $a = \frac{dv}{dt} = \frac{dv}{dr} \cdot \frac{dr}{dt} = v\frac{dv}{dr}$.

$\Rightarrow v\,dv = \mu r\,dr \Rightarrow v^2 = \mu r^2 + C$.

$v_0 = 0, r_0 = a \Rightarrow \mu a^2 + C = 0$

$\therefore C = -\mu a^2$.

令 $r = r_0 + S = 2a$.

$v = \mu 4a^2 - \mu a^2 = 3\mu a^2$

$\therefore v = \sqrt{3\mu} \cdot a$.

2.21 $mg - R = ma$ $\Rightarrow$ 代入 $R = kv^2$ 成正比故 $k > 0$.

$mg - kv^2 = ma$ $a = 0$ 时, $V_t = \sqrt{\frac{mg}{k}}$.

$mg - kv^2 = m\frac{dv}{dt}$. 又 $t = 0$ 时, $v = 0$

$\Rightarrow \frac{m\,dv}{mg - kv^2} = dt$ $\therefore C = 0$

$m\int \frac{dv}{mg - kv^2} = t + C$.

$\Rightarrow \frac{1}{2}\sqrt{\frac{k}{mg}} \ln\left|\frac{\sqrt{\frac{mg}{k}} + v}{\sqrt{\frac{mg}{k}} - v}\right| = \frac{k}{m}(t + C)$

$$\Rightarrow \boxed{\frac{\ln\left|\frac{V_t + v}{V_t - v}\right|}{V_t} = \frac{kt}{m}}$$

3. $F(t)=\begin{cases} 2t, & 0\le t\le 5s, \\ -5t+35, & 5s< t\le 7s \end{cases}$

$I=\int_0^7 F(t)dt = 35 N\cdot s$. 由 $I=m(V_t-V_0)$ 得 $V_t = 40 m/s$.

$0\sim 5s$, $a(t)=2t$, $V(t)=5+t^2$.

$x_1=\int_0^5 v(t)dt + x_0 = \frac{200}{3} m$

$5s\sim 7s$, $a(t)=-5t+35$, $V_5 = 30 m/s$

$v(t)=V_5+\int_5^t(-5t+35)dt = -\frac{5}{2}t^2+35t-\frac{75}{2}$.

$x_2=\int_5^7 v(t)dt = \frac{170}{3} m$

$\Delta x = \frac{370}{3} m$.

4. (1) $f=\mu F_N$. 由 $F_N = m\frac{v^2}{R}$, $f=-m\frac{dv}{dt}$.

$\Rightarrow -m\frac{dv}{dt}=\mu m\frac{v^2}{R}$. 得 $\frac{dv}{v^2}=-\frac{\mu dt}{R}$.

$\Rightarrow \frac{1}{v}=\frac{\mu t}{R}+C$. 又 $t=0$ 时, $v=v_0$ $\therefore C=\frac{1}{v_0}$.

$\Rightarrow \frac{1}{v}=\frac{\mu t}{R}+\frac{1}{v_0}$, $v=\frac{1}{\frac{\mu t}{R}+\frac{1}{v_0}}=v(t)$.

(2) 令 $v=\frac{v_0}{2}$, $\Rightarrow \frac{1}{v_0}=\frac{\mu t}{R}$ $\therefore t=\frac{R}{\mu v_0}$.

$S=\int_0^{\frac{R}{\mu v_0}} v(t)dt=\int_0^{\frac{R}{\mu v_0}}\frac{dt}{\frac{\mu t}{R}+\frac{1}{v_0}} = \frac{R}{\mu}\ln\left(\frac{\mu t}{R}+\frac{1}{v_0}\right)\Big|_0^{\frac{R}{\mu v_0}}$

$=\cancel{\frac{R}{\mu}\left(\ln\frac{2}{v_0}-\ln\frac{1}{v_0}\right)}=\frac{R}{\mu}\ln 2$.

5. $F = kmv^2$, $G = mg$.

上升过程，~~$F+G = -ma$~~ $F+G=-ma$.

$$kmv^2 + mg = -m\frac{dv}{dt} = -mv\frac{dv}{dx}.$$

即 $-\frac{v\,dv}{kv^2+g} = dx$，解得 $x = \frac{1}{2k}\ln\left(1+\frac{kv_0^2}{g}\right)$

(2) 返回地面时，$G - F = m\frac{dv}{dt}$

$$\Rightarrow mg - kmv^2 = m\frac{v\,dv}{dx}.$$

$$\Rightarrow \frac{v\,dv}{g-kv^2} = dx \Rightarrow v = \frac{v_0}{\sqrt{1+\frac{kv_0^2}{g}}}$$